

AyuPramana

IP-SAKTI Sahayak · SIH26045 · Ministry of Ayush (All India Institute of Ayurveda)
Voiceover script for the screen recording · 2 min 30 s · 354 words (about 140 words per minute)

How to read it

Read each block calmly and start it at the time shown. Pause for a breath between blocks. The script stands on its own: it tells the story from the problem to the solution, the innovation, and what comes next. Sections 2–5 are the evidence behind every claim — use them for questions from the judges.

1. Voiceover script

0:00 – 0:25 · The problem (61 words)

An Ayurvedic innovator who wants to protect and sell a product faces a legal maze: patents, geographical indications, trade marks, benefit sharing, drug licensing and advertising rules — in India and abroad. The rules are scattered, changing, and written for lawyers. So innovation stays unprotected, and India's traditional knowledge stays exposed. SIH26045 asks for a trusted, multilingual guide. That guide is AyuPramana.

0:25 – 0:54 · Our approach (72 words)

AyuPramana is a retrieval-augmented assistant that answers only from official law, never from memory. We ingest statutes and treaties from India Code, the WTO, the CBD and WIPO — thirteen instruments today, from the Patents and Biological Diversity Acts to the Nagoya Protocol and the 2024 WIPO treaty. Each is split along its own sections and versioned by date and file fingerprint. An agentic router sends every question to the right specialist.

0:54 – 1:20 · Specialists that cover the problem statement (63 words)

Five specialists cover the problem end to end: an IP adviser for patents, trade marks, GI, designs, copyright and plant varieties; a benefit-sharing compliance checklist; a traditional-knowledge prior-art guide that points to TKDL; a registry navigator; and a classifier that places a product as classical, proprietary, new drug, phytopharmaceutical, Ayurveda Aahar or cosmetic — using only the legal definitions it retrieves.

1:20 – 1:55 · Our innovation: trust you can verify (83 words)

Our core innovation is trust you can verify. Every sentence carries a citation — the section, its version date, the official link — and one click shows the exact legal text. A second AI pass fact-checks each sentence against its source and deletes anything unsupported, even “it isn't listed, so it's allowed” reasoning. A confidence score blends retrieval relevance, agreement between keyword and semantic search, and that fact-check. When the evidence is weak, AyuPramana refuses to guess and offers a human IP facilitator.

1:55 – 2:13 · Safety and languages (39 words)

Indian and international law run as separate pipelines, never mixed. Medical and dosage questions are declined, and personal data is masked before it reaches the model. Answers come in English, Hindi and Marathi through Bhashini, with every citation preserved.

2:13 – 2:30 · Next steps and close (36 words)

Next: the 2024 patent and biodiversity rules, the Ayurveda drug rules, FSSAI Aahar, and the PCT, Madrid and Hague systems; an expert-verified benchmark; and integration with IP facilitation centres. AyuPramana — every answer, with its pramāṇa.

2. Proof: how AyuPramana resolves SIH26045

Each requirement of the problem statement, and the feature that meets it.

What the problem statement asks for	How AyuPramana delivers it
Multilingual assistant	English, Hindi and Marathi interface and answers. Questions are searched in English; answers are translated back through Bhashini with an AI fallback. Citations are preserved and the official text is shown unchanged.
RAG-based and source-cited	Answers come only from retrieved official text. Every sentence ends with a citation (document, section, jurisdiction, version date, official link), and one click opens the exact source text.
Patents, GI, trade marks, copyright, designs, trade secrets, plant-variety rights	IP Routing specialist over the Patents, GI, Trade Marks, Designs, Copyright and PPVFR Acts plus TRIPS (including its provisions on undisclosed information).
Access-and-benefit-sharing duties (India's sovereignty over biological resources)	ABS Compliance specialist produces a cited checklist from the Biological Diversity Act, 2002 (with the 2023 amendment), the CBD and the Nagoya Protocol.
Is it a classical medicine, proprietary medicine, new drug, phytopharmaceutical, food or cosmetic?	Formulation Classifier: four quick-reply questions, then the category is decided only from retrieved legal definitions (Drugs and Cosmetics Act, including Chapter IVA on Ayurvedic drugs).

What the problem statement asks for	How AyuPramana delivers it
Traditional knowledge exposed to misappropriation; WIPO GR & ATK Treaty (2024)	Prior-art / TKDL pointer explains traditional knowledge as prior art from the Patents Act; the 2024 WIPO treaty's disclosure requirement is in the corpus; official search portals are linked once verified.
Fast-moving advertising and regulatory landscape	Drugs and Magic Remedies Act, 1954, including its Schedule. Example: "Can I advertise that my product cures diabetes?" is answered No, citing the Schedule entry for diabetes.
Recent changes (2024 patent and biodiversity rules) — guidance must stay current	Every file is version-tracked by hash and date. A changed document becomes a new version, old text is retired but kept for audit, and every citation shows the version date.
National and international regimes	India / International / Both toggle: two separate, strictly filtered pipelines, shown as side-by-side cards. The two are never mixed.
Plain-language guidance for practitioners, researchers, AYUSH startups, MSMEs, cultivators	Short plain answers, quick-reply buttons, starter questions, a mobile-friendly web app, and a "Talk to an IP facilitator" escalation with a reference number.
Dataset: TKDL, India Code, IP India, NBA	Nine Acts ingested from India Code; links to TKDL, IP India and NBA are kept in a team-verified registry list (never invented).

Against the four judging criteria

Criterion	How it is met
Answer accuracy	Grounded generation, sentence-level fact-check, on-topic check, and a guard against claims based on what a source does not mention.
Citation correctness	Invalid citations removed automatically; the evaluation harness measures citation precision and recall against expert-verified expectations.
Safe abstention	Confidence threshold, out-of-scope detection (medical dosage, unrelated topics), related provisions offered as pointers, and escalation to a human facilitator.
Multilingual quality	Citation markers verified after translation; per-language results in the evaluation report; quoted law always shown in its official language.

3. The innovation

- **Verified, sentence-level grounding:** a second AI pass deletes any sentence its cited source does not support, before the user sees the answer.
- **Absence-claim guard:** blocks the most dangerous legal error — "it isn't listed, so it's allowed" — caught while testing on the real Drugs and Magic Remedies Act.
- **Explainable confidence:** retrieval relevance + keyword/semantic agreement + fact-check, shown as High / Medium / Low.
- **Jurisdiction-separated pipelines:** India and international law answered independently, side by side.
- **Law-aware ingestion:** official PDFs split along their own Sections, Articles, Chapters and Schedules (footnotes, amendment markers and annexes handled), each chunk cited by its section number.
- **Version-tracked corpus:** file-hash versioning, so every answer shows which version of the law it used.
- **Privacy by design:** phone, email and Aadhaar-like numbers masked before the model or the log sees them; anonymous sessions only (DPDP Act, 2023).

4. Current status

- Working prototype: FastAPI backend, React web app, hybrid search (BM25 + multilingual embeddings), agentic router with five specialists, fact-checking, Bhashini/LLM translation, audit dashboard.
- Corpus: 13 official instruments, about 1,200 provision-level chunks — Patents, Biological Diversity, GI, Trade Marks, Designs, Copyright, PPVFR, Drugs and Cosmetics, and Drugs and Magic Remedies Acts; TRIPS, CBD, Nagoya Protocol and the WIPO GR & ATK Treaty (2024).
- Quality: 81 automated tests; live-tested with a hosted LLM in English, Hindi and Marathi.

5. Required next changes and future scope

- Add the remaining instruments: Patents Rules (2024 amendment), Biological Diversity Rules 2024 and ABS regulations, Drugs and Cosmetics Rules (Ayurveda, Siddha, Unani), FSSAI Ayurveda Aahar regulations, PCT, Madrid, Hague and Budapest.
- Have domain experts verify the evaluation questions and classifier wording; publish accuracy benchmarks.
- Verify the official portal links (TKDL, IP India, NBA, WIPO) so the registry navigator can show them.
- Activate Bhashini credentials; add more Indian languages and voice input.
- Host the public version (low-memory deployment is ready) and integrate with IP facilitation centres and AYUSH startup support.

All information given by AyuPramana is general information, not legal advice.